

Wheat — Crop Production Notes

Introduction and Local Names

The scientific name of wheat is *Triticum aestivum* L., and it belongs to the family Poaceae. In Urdu, wheat is called Gandum, in Punjabi it is known as Kanak, and in Pashto it is called Ghanum.

Crop Characteristics

Wheat is an annual, long-day crop with a determinate growth habit. It is self-pollinated, its seed is referred to as grain, and its pollination takes place on the spike.

History

According to tradition, wheat is associated with Hazrat Adam (A.S.) and the heavens, and with the Fertile Crescent and the Nile Delta. The first domestication of wheat is believed to have taken place in southeastern Turkey around 9,000 B.C., while pre-domesticated wheat dates back to around 23,000 B.C. The Fertile Crescent is a crescent-shaped region containing the comparatively moist and fertile land of otherwise arid and semi-arid Western Asia, along with the Nile Valley and Nile Delta of northeast Africa.

Top Wheat Producing Countries

According to FAO (2019) data, China led wheat production with 134,340,630 tonnes, followed by India with 98,510,000 tonnes, Russia with 85,863,132 tonnes, the USA with 47,370,880 tonnes, and France with 36,924,938 tonnes.

Wheat Production in Pakistan (2019)

In 2019, wheat was cultivated on an area of 9,200,000 hectares in Pakistan, with a total production of 27,000,000 tonnes and a yield of 2.93 tonnes per hectare. Pakistan ranked 8th in world wheat production that year.

Morphology

Wheat has a hypogeal mode of germination and a fibrous, adventitious root system. The root depth is normally about 60 cm, but it can extend up to 200 cm.

Shoot Development

The shoot develops from the plumule, and many tillers can arise from a single seed. The stem has pithy nodes and hollow internodes.

Spike

The wheat spike is made up of spikelets, each containing three florets, also called glumes. Each floret has two parts: the lemma and the palea. The lemma is the outer covering and contains the awn, while

the palea lies beneath the lemma and covers the grain.

Climate Requirements

Wheat is a winter, or Rabi, crop that requires low temperatures of about 4–10°C. Tillering takes place over about 1.5 months, from December to January. In snowy areas, wheat is sown either after the snow melts (April–May) or before snowfall (October–November).

Soil Requirements

Wheat grows best in loamy, well-drained clayey soils.

Primary Tillage

Primary tillage for wheat involves 4–6 ploughings and 2–3 plankings. The extent of tillage depends on the nature of the previous crop, the soil type, the soil condition, and the season.

Seed Bed Preparation

Seed bed preparation for wheat involves 2 cultivations and 2 plankings.

Sowing Time

For rainfed wheat, early sowing takes place from 15 to 31 October, mid sowing from November to December, and late sowing up to 15 January. For irrigated wheat, November is the optimum sowing time, with December as an alternative.

Dab Method

The Dab method is used for weed control. It involves achieving optimum moisture through a presoaking irrigation, followed by ploughing and planking, which buries the weeds in the soil and helps maintain soil fertility.

Seed Rate

The recommended seed rate for wheat is 100–120 kg per hectare. The exact rate depends on the sowing method, seed purity, germination percentage, and the variety being sown.

Sowing Method

Wheat can be sown by broadcasting, or through the Kera and Pora methods, as well as by using an automatic Rabi drill.

Fertilizer

For rainfed wheat, the recommended fertilizer dose is 85:58:58 kg per hectare of NPK. For irrigated wheat, the recommended dose is 115:65:65 kg per hectare of NPK. Half of the nitrogen is applied along with the full amount of phosphorus and potassium at sowing, and the remaining half of the nitrogen is

applied with the first irrigation. In sandy soils, nitrogen is applied in three splits: at sowing, and with the first and second irrigations.

Irrigation

Wheat generally requires 4–6 irrigations, totaling about 19 acre-inches of water. Irrigation is important at the tillering, booting, earing, milky, and grain formation stages.

Plant Protection Measures

Plant protection in wheat covers the management of weeds, insect pests, and diseases.

Wheat Varieties by Province

Punjab (Irrigated): Millat 2011, Punjab 2011, AAS, AARI 2011, Faisalabad 2008, Lasani 2008.

Punjab (Rainfed): NARC 2011, NARC 2009, BARS 2009, Chakwal 50.

Sindh (Irrigated): Sarsabaz, NIA Sunahri, NIA Amber, NIA Sundar, Imdad 2005, Mehran 89.

KPK: Pir Sabaq 2008, Sahar 2006, Fakhar-e-Khabar Pakhtun Khwa, Ata Habib, Amin 2010, Barsat, Kohat 2010.

Baluchistan: Zargoona, Zardana, Inqalab 91, Zarlashtha, Raskoh 2005, Lasani 2008, Sahar 2006.